Bayesian Inverse Problem

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1 Introduction

1.1 Inverse problem

References.

- Aster, Borchers, and Thurber, 2018 [pdf]
- Ito and Jin, 2015 [pdf]
- Kirsch, 2021 [pdf]
- **[W/IP]** Wikipedia/Inverse Problem.

Sketches.

1. **Definition. [W/IP]** "An *inverse problem* in science is the process of calculating from a set of observations the causal factors that produced them."
2. **Intuition.** "It is called an inverse problem because it starts with the effects and then calculates the causes.

   It is the inverse of a *forward problem*, which starts with the causes and then calculates the effects."
3. **Motivation.** "Inverse problems are some of the most important mathematical problems in science and mathematics because they tell us about parameters that we cannot directly observe."

4. **Application.** Inverse problems "have wide application in system identification, optics, radar, acoustics, communication theory, signal processing, medical imaging, computer vision, geophysics, oceanography, astronomy, remote sensing, natural language processing, machine learning, nondestructive testing, slope stability analysis and many other fields.

Communities.

- WIAS/Research Group 4: *Nonlinear Optimization & Inverse Problem*.

1.1.1 Forward problem vs. inverse problem

Physical Model. $\mathcal{G}(u) \to y$.

- $u \in X$ parameter vector/parameter function
- $\mathcal{G} : X \to Y$ forward response operator
- y results/observations
- Evaluation of $\mathcal{G}$ expensive

Forward Problem. Find the output y for given parameters u.
Find the *unknown data* $u \in X$ from *noisy observations*

$$y = \mathcal{G}(u) + \eta \text{ with } \eta \sim \mathcal{N}(0, \Gamma).$$

- $u \in X$ parameter vector/parameter function
- $\mathcal{G} : X \to Y$ forward response operator
- y result/observations
- Evaluation of $\mathcal{G}$ expensive.

Forward Problem. $\boxed{\text{Find the output } y \text{ for given parameters } u} \to$ **well-posed**.

Inverse Problem. $\boxed{\text{Find the parameters } u \text{ from (noisy) observations } y} \to$ **ill-posed**.

Example 1.1 (2D Inhomogeneous sideways heat equations)**.** *Consider the following 2D inhomogeneous sideways heat equation (SHE):*

$$\begin{cases} \partial_t u - \Delta u = f(t,x,y), & \text{in } [0,\infty) \times \mathbb{R}^2, \\ u(t,0,y) = \phi(t,y), & \text{in } [0,\infty) \times \mathbb{R}, \\ \partial_x u(t,0,y) = \psi(t,y) & \text{in } [0,\infty) \times \mathbb{R}, \end{cases} \tag{2D-iSHE}$$

where ϕ, ψ: temperature, flux data *given in the* accessible region $x = 0$, *f: heat source density.*

Approaches to represent solutions: separation of variables, transform methods, etc., see, e.g., Evans, 2010.

(i) ***Determination of temperature.*** *Denote the 2D Fourier transform of a function $f(t,y)$ by*

$$\hat{f}(\tau,\xi) = \mathcal{F}(f)(\tau,\xi) = \frac{1}{2\pi}\int_{\mathbb{R}^2} f(t,y)e^{-\mathrm{i}(\tau t + \xi y)}\mathrm{d}y\mathrm{d}t, \ \forall(\tau,\xi) \in \mathbb{R}^2. \tag{2D-Ft}$$

extend the domain of time t from $[0,\infty)$ to $\mathbb{R}$, then applying the 2D Fourier transform (2D-Ft) w.r.t. y, t to (2D-iSHE), obtain in the frequency space the following 2nd-order ODE:

$$\begin{cases} \partial_x^2 \hat{u} - (\mathrm{i}\tau + \xi^2)\hat{u}(\tau,x,\xi) = \hat{f}(\tau,x,\xi), & \text{in } \mathbb{R}^3, \\ \hat{u}(\tau,0,\xi) = \hat{\phi}(\tau,\xi), & \text{in } \mathbb{R}^2, \\ \partial_x \hat{u}(\tau,0,\xi) = \hat{\psi}(\tau,\xi), & \text{in } \mathbb{R}^2. \end{cases}$$

Solving this ODE yields

$$\hat{u}(\tau,x,\xi) = \hat{\phi}(\tau,\xi)\cosh(x\theta) + \hat{\psi}(\tau,\xi)\frac{\sinh(x\theta)}{\theta} - \int_0^x \frac{\sinh((x-\eta)\theta)}{\theta}\hat{f}(\tau,\eta,\xi)\mathrm{d}\eta, \ \forall(\tau,x,\xi) \in \mathbb{R}^3, \tag{$\hat{u}$}$$

where $\theta = \theta(\tau,\xi)$ is the principle value of $\sqrt{\mathrm{i}\tau + \xi^2}$ and is given by

$$\theta(\tau,\xi) = \sqrt{\frac{\sqrt{\tau^2+\xi^4}+\xi^2}{2}} + \mathrm{i}\,\mathrm{sgn}(\tau)\sqrt{\frac{\sqrt{\tau^2+\xi^4}-\xi^2}{2}}, \ \forall(\tau,\xi) \in \mathbb{R}^2.$$

This representation of $\hat{u}$ also indicates the ill-posedness of (2D-iSHE). Indeed, for $i = 1, 2$, consider u_i solves (2D-iSHE) with the data (f_i, ϕ_i, ψ_i), and then

$$\hat{u}_i(\tau, x, \xi) = \hat{\phi}_i(\tau, \xi)\cosh(x\theta) + \hat{\psi}_i(\tau, \xi)\frac{\sinh(x\theta)}{\theta} - \int_0^x \frac{\sinh((x-\eta)\theta)}{\theta}\hat{f}_i(\tau, \eta, \xi)\mathrm{d}\eta,\ \forall(\tau, x, \xi) \in \mathbb{R}^3,\ \textit{for } i = 1, 2.$$

Subtract these, term by term, to obtain:

$$(\hat{u}_1 - \hat{u}_2)(\tau, x, \xi) = (\hat{\phi}_1 - \hat{\phi}_2)(\tau, \xi)\cosh(x\theta) + (\hat{\psi}_1 - \hat{\psi}_2)(\tau, \xi)\frac{\sinh(x\theta)}{\theta} - \int_0^x \frac{\sinh((x-\eta)\theta)}{\theta}(\hat{f}_1 - \hat{f}_2)(\tau, \eta, \xi)\mathrm{d}\eta,\ \forall(\tau, x, \xi) \in \mathbb{R}^3.$$

The perturbations in the data, which are indicated by $(\hat{f}_1 - \hat{f}_2, \hat{\phi}_1 - \hat{\phi}_2, \hat{\psi}_1 - \hat{\psi}_2)$, are magnified by the hyperbolic function cosh *and even a singular kernel $\frac{\sinh(x\theta)}{\theta}$ whose a singularity is the origin of the frequency space, i.e., $(\tau, \xi) = (0, 0)$.*

Remark 1.1 (Generalization: $x = 0$ to $x = x_0 \in \mathbb{R}$)**.** *Note that by using the 2D Fourier transform (2D-Ft), the solution of (2D-iSHE) where the Cauchy data (ϕ, ψ) is posed at $x = x_0 \in \mathbb{R}$ instead of $x = 0$, can be represented as*

$$\hat{u}(\tau, x, \xi) = \hat{\phi}(\tau, \xi)\cosh((x - x_0)\theta) + \hat{\psi}(\tau, \xi)\frac{\sinh((x - x_0)\theta)}{\theta} - \int_{x_0}^x \frac{\sinh((x-\eta)\theta)}{\theta}\hat{f}(\tau, \eta, \xi)\mathrm{d}\eta,\ \forall(\tau, x, \xi) \in \mathbb{R}^3.$$

- ***Ill-posedness.*** *Both kernels* $\cosh(x\theta)$, $\frac{\sinh(x\theta)}{\theta}$ *are unbounded for arbitrarily $x \neq 0$ when $\|(\tau, \xi)\|_2^2 = \tau^2 + \xi^2 \to \infty$, and thus small errors in the data can blow up and ultimately destroy the solution.*
- ***Measured data.*** *The measured data, denoted by $(f_\delta, \phi_\delta, \psi_\delta)$, where $\delta > 0$ is called an* error level *(see, e.g., Ito and Jin, 2015; Kirsch, 2021), is typically assumed to satisfy*

$$\max\{\|f_\delta - f\|_{L^2(\mathbb{R}^3)}, \|\phi_\delta - \phi\|_{L^2(\mathbb{R}^2)}, \|\psi_\delta - \psi\|_{L^2(\mathbb{R}^2)}\} \leq \delta. \tag{noise}$$

(ii) ***Determination of the thermal flux.*** *Differentiate the RHS of ($\hat{u}$) w.r.t. x, with the help of Leibnitz integral rule for the integral term, to obtain the following representation of the thermal flux w.r.t. x*

$$\partial_x\hat{u}(\tau, x, \xi) = \theta\hat{\phi}(\tau, \xi)\sinh(x\theta) + \hat{\psi}(\tau, \xi)\cosh(x\theta) - \int_0^x \cosh((x-\eta)\theta)\hat{f}(\tau, \eta, \xi)\mathrm{d}\eta,\ \forall(\tau, x, \xi) \in \mathbb{R}^3.$$

Blow-up kernels: $\sinh(x\theta)$, $\cosh((x-\eta)\theta) \to$ *ill-posedness.*

(iii) ***Numerical implementation.*** *The Cauchy data $(u, \partial_x u)|_{x=0} = (\phi, \psi)$ and the source term $f(t, x, y)$ are given and sampled at an equidistant grid in the domain $[L, R]^2$ with $n \times n$ points (n: sampling frequency), and the discretizations of (f, ϕ, ψ) are denoted by $(\mathbf{F}, \Phi, \Psi)$, to distinguish them and their continuous counterparts.*

Main tools: MATLAB *builtin-in function* `fft2`*, which is used to carry out the 2D fast Fourier transforms (abbr., 2D FFT) of the given data, and* `ifft2`*, which is used to obtain 2D inverse fast Fourier transforms (abbr., 2D IIFT).*

Denote $\bar{x} \in [0, 1]$ the point at which we choose to observe the solutions and the thermal fluxes. Use Gauss-Legendre quadrature of order m for numerical integrations, where $m \in \mathbb{Z}_{>0}$ is chosen by users, i.e.,

$$\int_0^{\bar{x}} \widehat{f}_\delta(\tau, \eta, \xi)\frac{\sinh((\bar{x} - \eta))}{\theta}\mathrm{d}\eta \approx \sum_{k=1}^m w_{m,k}\widehat{f}_\delta(\tau, x_{m,k}, \xi)\frac{\sinh((\bar{x} - x_{m,k})\theta)}{\theta},$$
$$\int_0^{\bar{x}} \widehat{f}_\delta(\tau, \eta, \xi)\cosh((\bar{x} - \eta)\theta)\mathrm{d}\eta \approx \sum_{k=1}^m w_{m,k}\widehat{f}_\delta(\tau, x_{m,k}, \xi)\cosh((\bar{x} - x_{m,k})\theta),$$

where $x_{m,k}$ and $w_{m,k}$ are the nodes and weights of the Gauss-Legendre quadrature of degree m for the interval $[0, \bar{x}]$, resp. To apply these numerical integration formulas, it is necessary to carry out the 2D FFT of f_δ at each Gauss-Legendre quadrature's node.

Add random noises to the (discrete) exact input data $(\mathbf{F}, \Phi, \Psi)$, i.e.:

$$\begin{cases} \mathbf{F}_\delta(\cdot, x_{m,k}, \cdot) = \mathbf{F}(\cdot, x_{m,k}, \cdot) + \delta\,\mathtt{randn}(\mathtt{size}(\mathbf{F}(\cdot, x_{m,k}, \cdot))),\ \forall k = 1, \ldots, m, \\ \Phi_\delta = \Phi + \delta\,\mathtt{randn}(\mathtt{size}(\Phi)),\ \Psi_\delta = \Psi + \delta\,\mathtt{randn}(\mathtt{size}(\Psi)), \end{cases} \tag{noise data}$$

where `randn()` *is a* MATLAB *built-in function.*

Denote $\|\cdot\|_{\overline{\mathrm{F}}}$ the "average" Frobenius norm, i.e.:

$$\|A\|_{\overline{\mathrm{F}}} := \frac{1}{\sqrt{mn}}\|A\|_{\mathrm{F}} = \frac{1}{\sqrt{mn}}\left(\sum_{i=1}^m\sum_{j=1}^n |A_{ij}|^2\right)^{\frac{1}{2}},\ \forall A \in \mathrm{Mat}_{\mathbb{C}}(m \times n),$$

where $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm, we can define the total noises of the 3D array $\mathbf{F}_\delta$ and the matrices $(\Phi_\delta, \Psi_\delta)$ as

$$\mathrm{TN}(\mathbf{F}, \mathbf{F}_\delta) := \frac{1}{n\sqrt{m}} \left(\sum_{k=1}^{m} \|(\mathbf{F}_\delta - \mathbf{F})(\cdot, x_{m,k}, \cdot)\|_{\mathrm{F}}^2 \right)^{\frac{1}{2}}, \ \mathrm{TN}(\Phi, \Phi_\delta) := \|\Phi_\delta - \Phi\|_{\overline{\mathrm{F}}}, \ \mathrm{TN}(\Psi, \Psi_\delta) := \|\Psi_\delta - \Psi\|_{\overline{\mathrm{F}}}, \ \forall \delta > 0. \tag{total noise}$$

After doing this, we obtain the noisy data $(\mathbf{F}_\delta, \Phi_\delta, \Psi_\delta)$ satisfying the condition $\max\{\mathrm{TN}(\mathbf{F}, \mathbf{F}_\delta), \mathrm{TN}(\Phi, \Phi_\delta), \mathrm{TN}(\Psi, \Psi_\delta)\} \le \delta$, which can be interpreted as the discrete analogue of (noise). We also use the norm $\|\cdot\|_{\overline{\mathrm{F}}}$ to calculate root mean square and relative errors, e.g.,

$$\mathrm{RMSE}(\mathbf{U}_{\mathcal{R}}^\delta) := \|\mathbf{U}_{\mathcal{R}}^\delta - \mathbf{U}_{\mathrm{exact}}\|_{\overline{\mathrm{F}}}, \ \mathrm{RE}(\mathbf{U}_{\mathcal{R}}^\delta) := \frac{\|\mathbf{U}_{\mathcal{R}}^\delta - \mathbf{U}_{\mathrm{exact}}\|_{\overline{\mathrm{F}}}}{\|\mathbf{U}_{\mathrm{exact}}\|_{\overline{\mathrm{F}}}},$$

where $\mathbf{U}_{\mathrm{exact}}$, $\mathbf{U}_{\mathcal{R}}^\delta$ is the discrete version of the solution u of (2D-iSHE) and its regularized analogue $u_{\mathcal{R}}^\delta$ on the sampled grid.

Algorithm 1 Numerical implementation of a regularization method proposed for (2D-iSHE).

1: **Inputs:** observed point $x_{\mathrm{obs}} \in \mathbb{R}$, Cauchy data $[\phi, \psi]$ at $x = 0$, source term f, boundaries L and R of the computational domain $\Omega_{L,R}$, sampling frequency n, order $m \in \mathbb{Z}_{>0}$ of Gauss-Legendre quadrature, (approximate) nodes $x_{k,m}$ and (approximate) weights $w_{k,m}$ of the Gauss-Legendre quadrature of degree m for the interval $[0, x_{\mathrm{obs}}]$, noisy level $\delta > 0$, exact solution u or the set of its values at the grid points;
2: Generate an equidistant grid in the computational domain $\Omega_{L,R}$; ▷ Treat the origin $(0, 0)$ carefully
3: Compute the discrete exact data $(\mathbf{F}, \Phi, \Psi)$;
4: Generate the discrete noisy data $(\mathbf{F}_\delta, \Phi_\delta, \Psi_\delta)$ by (noise data);
5: (Optional) Compute the triple of the averaged noises $(\mathrm{TN}(\mathbf{F}, \mathbf{F}_\delta), \mathrm{TN}(\Phi, \Phi_\delta), \mathrm{TN}(\Psi, \Psi_\delta))$ of the discrete noisy data generated in Step 4 by (total noise); ▷ This triple will slightly change in magnitude in each execution due to randomness.
6: Compute 2D FFT $(\widehat{\mathbf{F}_\delta}, \widehat{\Phi_\delta}, \widehat{\Psi_\delta})$ of the discrete noisy data in Step 4 by `fft2`;
7: Compute 2D FFT of the discrete versions $\widehat{\mathbf{U}_\delta}$ and $\widehat{\mathbf{U}_{\mathcal{R}}^\delta}$ of $(\hat{u})$ and the regularized one(s);
8: Carry out both the discrete versions $\mathbf{U}_\delta$ and $\mathbf{U}_{\mathcal{R}}^\delta$ of the non-regularized and regularized solutions u_δ and $u_{\mathcal{R}}^\delta$ by `ifft2`;
9: Compute the corresponding Root Mean Square Error $\mathrm{RMSE}(\mathbf{U}_{\mathcal{R}}^\delta)$ and Relative Error $\mathrm{RE}(\mathbf{U}_{\mathcal{R}}^\delta)$;
10: (Optional) 3D-Plot the discrete non-regularized solution $\mathbf{U}_\delta$ and regularized solution $\mathbf{U}_{\mathcal{R}}^\delta$ on the equidistant grid generated in Step 2;
11: **Output:** Triple of the averaged noises $(\mathrm{TN}(\mathbf{F}, \mathbf{F}_\delta), \mathrm{TN}(\Phi, \Phi_\delta), \mathrm{TN}(\Psi, \Psi_\delta))$ (optional), non-regularized discrete solution $\mathbf{U}_\delta$, regularized discrete solution $\mathbf{U}_{\mathcal{R}}^\delta$, errors $\mathrm{RMSE}(\mathbf{U}_{\mathcal{R}}^\delta)$, $\mathrm{RE}(\mathbf{U}_{\mathcal{R}}^\delta)$, 3D plots of non-regularized and regularized solutions.

The MATLAB scripts and a gallery for this algorithm for (2D-iSHE) can be downloaded at the following URLs:

- NQBH/math/seminar/Bayesian inverse problem/MATLAB
- NQBH/math/seminar/Bayesian inverse problem/gallery.

1.1.2 Deterministic Optimization Problem

Find the unknown data $u \in X$ from noisy observations

$$y = \mathcal{G}(u) + \eta.$$

Deterministic optimization problem

$$\min_{u \in X} \frac{1}{2} \|y - \mathcal{G}(u)\|^2 + R(u).$$

- $\|y - \mathcal{G}(u)\|$ potential/data misfit
- R regularization term
- Large-scale, deterministic optimization problem
- No quantification of the uncertainty in the unknown u
- Proper choice of the regularization term R

1.2 Uncertainty quantification

References.

- **[W/U]** Wikipedia/Uncertainty.
- **[W/UQ]** Wikipedia/Uncertainty Quantification.

Sketches.

1. **Definitions.** **[W/U]** "*Uncertainty* refers to epistemic situations involving imperfect or unknown information."

   [W/UQ] "*Uncertainty quantification* (UQ) is the science of quantitative characterization and reduction of uncertainties in both computational and real world applications."

2. **Intuition.** Uncertainty quantification "tries to determine how likely certain outcomes are if some aspects of the system are not exactly known".

3. **Motivation.** "Many problems in the natural sciences and engineering are also rife with sources of uncertainty." E.g., (see, e.g., **[Schillings_Teckentrup2017]**, Slides 3–4) *environmental systems* (weather, climate, seismic, subsurface geophysics), *engineering systems* (automobiles, aircraft, bridges, structures), *biological systems* (health & medicine, pharmaceuticals, gene expression, cancer research), *physical systems* (nano-optics, quantum physics, radioactive decay), etc.

4. **Sources of uncertainty.** **[W/UQ]** "Uncertainty can enter mathematical models and experimental measurements in various contexts."

   Categorization of sources of uncertainty. Consider: parameter uncertainty, parametric variability, structural uncertainty/model inadequacy/model bias/model discrepancy, algorithmic uncertainty/numerical uncertainty/discrete uncertainty, experimental uncertainty/observation error, interpolation uncertainty.

   Classification of uncertainty.

   - *Aleatoric uncertainty/statistical uncertainty* is representative of unknowns that differ each time we run the same experiment. E.g., MATLAB built-in function `rand()`.

     "The quantification for the aleatoric uncertainties can be relatively straightforward, where traditional (frequentist) probability is the most basic form."

   - *Epistemic uncertainty/systematic uncertainty* is due to things one could in principle know but do not in practice, which may be because a measurement is not accurate, because the model neglects certain effects, or because particular data has been deliberately hidden. E.g., $g = 9.8\text{m/s}^2$ ignores the effects of air resistance.

     "To evaluate epistemic uncertainties, the efforts are made to understand the (lack of) knowledge of the system, process or mechanism."

   "Uncertainty quantification intends to explicitly express both types of uncertainty separately."

   Mathematical perspective of aleatoric & epistemic uncertainty. "In mathematics, uncertainty is often characterized in terms of a probability distribution. From that perspective, epistemic uncertainty means not being certain what the relevant probability distribution is, and aleatoric uncertainty means not being certain what a random sample drawn from a probability distribution will be."

5. **Quantification & minimization of uncertainties.**

   - Uncertainty quantification in numerical simulations
   - Minimization of uncertainties in design & control problems
   - Identification of uncertain parameters from noisy observations

   Related research fields. Numerical analysis, methods for optimal control/optimization problems, approximation theory, stochastic, statistics, algorithms & data structures.

Communities.

- WIAS/Research Group 5: *Interacting Random Systems*.
- WIAS/Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*.

2 Bayesian Inverse Problems

References.

- **[Schillings_Teckentrup2017]** Schillings, Claudia & Teckentrup, Aretha, 2017. *Mathematical Foundations of Bayesian Inverse Problems: Well-posedness & Statistical Estimates in Bayesian Inverse Problems.* [pdf]

In Dashti and Stuart, 2017, the authors highlight the mathematical and computational structure relating to the formulation of, and development of algorithms for, the Bayesian approach to inverse problems in differential equations, which is a fundamental approach in the quantification of uncertainty within applications involving the blending of mathematical models with data.

Find the unknown data $u \in X$ from noisy observations $y = \mathcal{G}(u) + \eta$.

Bayesian inverse problem

- u, η, y random variables/fields
- Prior μ_0, posterior μ^y ???
- Goal of computation: moments of system quantities under the posterior w.r.t. noisy data

Prior $\to$ Forward Model $G \to$ (Noisy) Data y.

$$\text{Posterior} \to \text{Forward Model } G \to \begin{cases} \to \text{(Noisy) Data } y \to \text{Posterior}, \\ \to \text{Quantity of interest.} \end{cases}$$

- Quantification of uncertainty in u & system quantities
- Well-posedness of the inverse problem ???
- Incorporation of prior knowledge on the uncertain data u
- Need for efficient approximations of the posterior

2.1 Bayesian Inversion in $\mathbb{R}^n$

Consider the problem of finding $\mathbf{u} \in \mathbb{R}^n$ from $\mathbf{y} \in \mathbb{R}^J$ where $\mathbf{u}$ and $\mathbf{y}$ are related by the equation $\mathbf{y} = G(\mathbf{u})$.

We refer to $\mathbf{y}$ as *observed data* and to $\mathbf{u}$ as the *unknown*.

This problem may be difficult for a number of reasons.

The authors highlight 2 of these, both particularly relevant to their future developments.

1. The 1st difficulty, which may be illustrated in the case where $n = J$, concerns the fact that often the equation is perturbed by noise and so we should really consider the equation

   $$\mathbf{y} = G(\mathbf{u}) + \boldsymbol{\eta},$$

   where $\boldsymbol{\eta} \in \mathbb{R}^J$ represents the *observational noise* which enters the observed data.

   Assume further that G maps $\mathbb{R}^J$ into a proper subset of itself, Im_G, and that G has a unique inverse as a map from Im_G into $\mathbb{R}^J$.

   It may then be the case that, because of the noise, $\mathbf{y} \notin \mathrm{Im}_G$ so that simply inverting G on the data $\mathbf{y}$ will not be possible.

   Furthermore, the specific instance of $\boldsymbol{\eta}$ which enters the data may not be known to us; typically, at best, only the statistical properties of a typical noise $\boldsymbol{\eta}$ are known.

   Thus we cannot subtract $\boldsymbol{\eta}$ from the observed data $\mathbf{y}$ to obtain something in Im_G.

   Even if $\mathbf{y} \in \mathrm{Im}_G$, the uncertainty caused by the presence of noise $\boldsymbol{\eta}$ causes problems for the inversion.

2. The 2nd difficulty is manifest in the case where $n > J$ so that the system is *underdetermined*: the number of equations is smaller than the number of unknowns.

   How do we attach a sensible meaning to the concept of solution in this case where, generically, there will be many solutions?

Thinking probabilistically enables us to overcome both of these difficulties.

We will treat $\mathbf{u}, \mathbf{y}$, and $\boldsymbol{\eta}$ as random variables and determine the joint probability distribution of $(\mathbf{u}, \mathbf{y})$.

We then define the "solution" of the inverse problem to be the probability distribution of $\mathbf{u}$ given $\mathbf{y}$, denoted $\mathbf{u}|\mathbf{y}$.

This allows us to $\boxed{\text{model the noise via its statistical properties}}$, even if we do not know the exact instance of the noise entering the given data.

And it also allows us to specify a priori the form of solutions that we believe to be more likely, thereby enabling us to attach weights to multiple solutions which explain the data.

This is the *Bayesian approach* to inverse problems.

To this end, we define a random variable $(\mathbf{u}, \mathbf{y}) \in \mathbb{R}^n \times \mathbb{R}^J$ as follows.

We let $\mathbf{u} \in \mathbb{R}^n$ be a random variable with (Lebesgue) density $\rho_0(\mathbf{u})$.

Assume that $\mathbf{y}|\mathbf{u}$ ($\mathbf{y}$ given $\mathbf{u}$) is defined via the formula (**??**) where $G : \mathbb{R}^n \to \mathbb{R}^J$ is measurable and $\boldsymbol{\eta}$ is independent of $\mathbf{u}$ (we sometimes write this as $\boldsymbol{\eta} \perp \mathbf{u}$) and distributed according to measure $\mathbb{Q}_0$ with Lebesgue density $\rho(\boldsymbol{\eta})$.

Then $\mathbf{y}|\mathbf{u}$ is simply found by shifting $\mathbb{Q}_0$ by $G(\mathbf{u})$ to measure $\mathbb{Q}_{\mathbf{u}}$ with Lebesgue density $\rho(\mathbf{y} - G(\mathbf{u}))$.

It follows that $(\mathbf{u}, \mathbf{y}) \in \mathbb{R}^n \times \mathbb{R}^J$ is a random variable with Lebesgue density $\rho(\mathbf{y} - G(\mathbf{u}))\rho_0(\mathbf{u})$.

The following theorem allows us to calculate the distribution of the random variable $\mathbf{u}|\mathbf{y}$:

Theorem 2.1 (Bayes' Theorem)**.** *Assume that*

$$Z := \int_{\mathbb{R}^n} \rho(\mathbf{y} - G(\mathbf{u}))\rho_0(\mathbf{u})\mathrm{d}\mathbf{u} > 0.$$

Then $\mathbf{u}|\mathbf{y}$ is a random variable with Lebesgue density $\rho^{\mathbf{y}}(\mathbf{u})$ given by

$$\rho^{\mathbf{y}}(\mathbf{u}) = \frac{1}{Z}\rho(\mathbf{y} - G(\mathbf{u}))\rho_0(\mathbf{u}).$$

Remark 2.1. *The following remarks establish the nomenclature[1] of Bayesian statistics and also frame the previous theorem in a manner which generalizes to the infinite-dimensional setting.*

- *$\rho_0(\mathbf{u})$ is the **prior density**.*
- *$\rho(\mathbf{y} - G(\mathbf{u}))$ is the **likelihood**.*
- *$\rho^{\mathbf{y}}(\mathbf{u})$ is the **posterior density**.*
- *It will be useful in what follows to define*

  $$\Phi(\mathbf{u}; \mathbf{y}) = -\log \rho(\mathbf{y} - G(\mathbf{u})).$$

  *We call Φ the **potential**.*

  *This is the **negative log likelihood**.*
- *Note that Z is the probability of $\mathbf{y}$.*

  Bayes' formula expresses

  $$\mathbb{P}(\mathbf{u}|\mathbf{y}) = \frac{1}{\mathbb{P}(\mathbf{y})}\mathbb{P}(\mathbf{y}|\mathbf{u})\mathbb{P}(\mathbf{u}).$$
- *Let $\mu^{\mathbf{y}}$ be a measure on $\mathbb{R}^n$ with density $\rho^{\mathbf{y}}$ and μ_0 a measure on $\mathbb{R}^n$ with density ρ_0.*

  Then the conclusion of Theorem 2.1 may be written as:[2]

  $$\frac{\mathrm{d}\mu^{\mathbf{y}}}{\mathrm{d}\mu_0}(\mathbf{u}) = \frac{1}{Z}\exp(-\Phi(\mathbf{u}; \mathbf{y})), \;\; Z = \int_{\mathbb{R}^n} \exp(-\Phi(\mathbf{u}; \mathbf{y}))\mu_0(\mathrm{d}\mathbf{u}).$$

[1] **nomenclaure**: [n] a system of naming things, especially in a branch of science.

[2] Indeed, since $\mu^{\mathbf{y}}$, μ_0 are measures on $\mathbb{R}^n$ with densities $\rho^{\mathbf{y}}$,ρ_0, respectively, we have $\mathrm{d}\mu^{\mathbf{y}}(\mathbf{u}) = \rho^{\mathbf{y}}\mathrm{d}\mathbf{u}$, $\mathrm{d}\mu_0(\mathbf{u}) = \rho_0(\mathbf{u})\mathrm{d}\mathbf{u}$, and thus the LHS can be written as

$$\frac{\mathrm{d}\mu^{\mathbf{y}}}{\mathrm{d}\mu_0}(\mathbf{u}) = \frac{\rho^{\mathbf{y}}(\mathbf{u})\mathrm{d}\mathbf{u}}{\rho_0(\mathbf{u})\mathrm{d}\mathbf{u}} = \frac{\rho^{\mathbf{y}}(\mathbf{u})}{\rho_0(\mathbf{u})}.$$

By definition of the potential Φ, we have

$$Z = \int_{\mathbb{R}^n} \exp(-\Phi(\mathbf{u}; \mathbf{y}))\mu_0(\mathrm{d}\mathbf{u}) = \int_{\mathbb{R}^n} \exp(\log \rho(\mathbf{y} - G(\mathbf{u})))\mu_0(\mathrm{d}\mathbf{u}) = \int_{\mathbb{R}^n} \rho(\mathbf{y} - G(\mathbf{u}))\rho_0(\mathbf{u})\mathrm{d}\mathbf{u},$$

which is assumed to be positive to make sense of the denominator. The RHS can be written as

$$\frac{1}{Z}\exp(-\Phi(\mathbf{u}; \mathbf{y})) = \frac{1}{Z}\exp(\log \rho(\mathbf{y} - G(\mathbf{y}))) = \frac{1}{Z}\rho(\mathbf{y} - G(\mathbf{u})).$$

Multiplying both sides with $\rho_0(\mathbf{u})$ yields the desired equality in Theorem 2.1.

Thus $\boxed{\textit{the posterior is absolutely continuous w.r.t. the prior}}$, *and* the Radon-Nikodym derivative is proportional to the likelihood.

This is rewriting Bayes' formula in the form

$$\frac{1}{\mathbb{P}(\mathbf{u})}\mathbb{P}(\mathbf{u}|\mathbf{y}) = \frac{1}{\mathbb{P}(\mathbf{y})}\mathbb{P}(\mathbf{y}|\mathbf{u}).$$

- *The expression for the Radon-Nikodym derivative is to be interpreted as the statement that, for all measurable* $f : \mathbb{R}^n \to \mathbb{R}$,

$$\mathbb{E}^{\mu^{\mathbf{y}}} f(\mathbf{u}) = \mathbb{E}^{\mu_0}\left(\frac{\mathrm{d}\mu^{\mathbf{y}}}{\mathrm{d}\mu_0}(\mathbf{u})f(\mathbf{u})\right).$$

Alternatively we may write this in integral form as

$$\int_{\mathbb{R}^n} f(\mathbf{u})\mu^{\mathbf{y}}(\mathrm{d}\mathbf{u}) = \int_{\mathbb{R}^n}\left(\frac{1}{Z}\exp(-\Phi(\mathbf{u};\mathbf{y}))f(\mathbf{u})\right)\mu_0(\mathrm{d}\mathbf{u}) = \frac{\int_{\mathbb{R}^n}\exp(-\Phi(\mathbf{u};\mathbf{y}))f(\mathbf{u})\mu_0(\mathrm{d}\mathbf{u})}{\int_{\mathbb{R}^n}\exp(-\Phi(\mathbf{u};\mathbf{y}))\mu_0(\mathrm{d}\mathbf{u})}.$$

2.2 Inverse Heat Equation

This inverse problem illustrates the 1st difficulty, which motivates the Bayesian approach to inverse problems.

Let $D \subset \mathbb{R}^d$ be a bounded open set, with Lipschitz boundary ∂D.

Then define the Hilbert space H and operator A as follows:

$$H = (L^2(D), \langle\cdot,\cdot\rangle, \|\cdot\|);\ A = -\Delta,\ \mathcal{D}(A) = H^2(D) \cap H_0^1(D).$$

We make the following assumption about the spectrum of A which is easily verified for simple geometries, but in fact holds quite generally.

Assumption 2.1. *The eigenvalue problem* $A\varphi_j = \alpha_j\varphi_j$ *has a countably infinite set of solutions, indexed by* $j \in \mathbb{Z}^+$.

They may be normalized to satisfy the L^2*-orthonormality condition*

$$\langle\varphi_j, \varphi_k\rangle = \delta_{jk} := \begin{cases} 1, & j = k, \\ 0, & j \neq k, \end{cases}$$

and form a basis for H.

Furthermore, the eigenvalues are positive and, if ordered to be increasing, satisfy $\alpha_j \asymp j^{2/d}$.

The notation $\asymp$ (i.e., "asymptotic to") denotes the existence of constants $C^{\pm} > 0$ s.t.

$$C^- j^{2/d} \leq \alpha_j \leq C^+ j^{2/d},\ \forall j \in \mathbb{N}.$$

Example 2.1 (1D Gaussian, linear example)**.** *Forward response operator* $g \in L(\mathbb{R}, \mathbb{R})$
Prior:

$$\mu_0 = \mathcal{N}(0, \sigma_0^2),\ \sigma_0 \in \mathbb{R},\ \textit{Leb. dens.}\ \rho_0(u) = \frac{1}{\sigma_0\sqrt{2\pi}} \exp\left(-\frac{\|u\|^2}{\sigma_0^2}\right).$$

Noise:

$$\eta \sim \mathcal{N}(0, \gamma^2),\ \gamma \in \mathbb{R},\ \textit{Leb. dens.}\ \rho_0(\eta) = \frac{1}{\gamma\sqrt{2\pi}} \exp\left(-\frac{\|\eta\|^2}{2\gamma^2}\right).$$

Bayes' formula

$$\rho(u|y) = \frac{\rho(y|u)\rho(u)}{\int_{\mathbb{R}^2} \rho(y|u)\rho(u)\mathrm{d}u}.$$

Completing the square

$$\mu^y = \mathcal{N}\left(\frac{\sigma_0^2 g}{\gamma^2 + g^2\sigma_0^2} y, \sigma_0^2 - \frac{\sigma_0^4 g^2}{\gamma^2 + g^2\sigma_0^2}\right).$$

By assumption, we have for the prior density $\rho_0(u)$

$$\rho_0(u) \propto \exp\left(-\frac{1}{2\sigma_0^2} u^2\right),$$

and the noise has density

$$\rho(\eta) \propto \exp\left(-\frac{1}{2\gamma^2}\eta^2\right).$$

Thus, by Bayes' formula, the posterior is given by

$$\rho^y(u) \propto \exp\left(-\frac{1}{2\sigma_0^2}u^2 - \frac{1}{2\gamma^2}(y - gu)^2\right) = \exp\left(-\frac{1}{2}\left(\frac{1}{\sigma_0^2} + \frac{g^2}{\gamma^2}\right)u^2 - 2\frac{gy}{\gamma^2}u + \frac{y^2}{u^2}\right).$$

Defining

$$a = \frac{1}{\sigma_0^2} + \frac{g^2}{\gamma^2},\ b = \frac{gy}{\gamma^2},\ c = \frac{y^2}{\gamma^2},$$

we are interested in constants m, K, σ*, s.t.*

$$\rho^\gamma(u) \propto \exp\left(-\frac{1}{2\sigma^2}(u - m)^2 + K\right).$$

By completing the square, we obtain

$$au^2 - 2bu + c = a\left(u^2 - 2\frac{b}{a}u + \frac{c}{a}\right) = a\left(u - \frac{b}{a}\right)^2 + c - \frac{b^2}{a},$$

and thus

$$\sigma = \frac{1}{a} = \frac{\sigma_0^2\gamma^2}{\gamma^2 + g^2\sigma_0^2},\ m = \frac{b}{a} = \frac{\sigma_0^2 g}{\gamma^2 + \sigma_0^2 g^2} y,$$

and for the constant K*, which does not depend on* u*, we obtain*

$$K = c - \frac{b^2}{a} = \frac{y^2}{\gamma^2} - \frac{\sigma_0^2 g^2 y^2}{y^4 + y^2 g^2 \sigma^2}.$$

Hence,

$$\rho^y(u) \propto \exp\left(-\frac{1}{2\left(\frac{\sigma_0^2\gamma^2}{\gamma^2 + g^2\sigma_0^2}\right)^2}\left(u - \frac{\sigma_0^2 g}{\gamma^2 + g^2\sigma_0^2} y\right)^2\right).$$

2.3 Estimators

- Maximum a posteriori estimate (MAP)

$$u_{\mathrm{MAP}} = \arg\max_{u\in\mathbb{R}^n} \rho^y(u).$$

- Conditional mean (CM)

$$u_{\mathrm{CM}} = \mathbb{E}(u|y) = \int_{\mathbb{R}^n} u\rho^y(u)\mathrm{d}u.$$

- Maximum likelihood estimate (ML)

$$u_{\mathrm{ML}} = \arg\max_{u\in\mathbb{R}^n} \rho(y-\mathcal{G}(u)).$$

- Conditional variance (speed estimator)

$$\mathbb{V}(u|y) = \int_{\mathbb{R}^n} (u-u_{\mathrm{CM}})(u-u_{\mathrm{CM}})^\top \rho^y(u)\mathrm{d}u.$$

Example 2.2 (Linear Gaussian Case)**.** ***Connection of MAP & CM & optimization.***
Forward response operator $A \in L(\mathbb{R}^n, \mathbb{R}^J)$
Prior: $\mu_0 = \mathcal{N}(m_0, C_0)$
Noise: $\eta \sim \mathcal{N}(0,\Gamma)$
Posterior:

$$\mu^y = \mathcal{N}(m,C) \text{ with } m = m_0 + C_0A^*(AC_0A^*+\Gamma)^{-1}(y-Am_0) \text{ and } C = C_0 - C_0A^*(AC_0A^*+\Gamma)^{-1}AC_0.$$

$$\begin{aligned} u_{\mathrm{MAP}} &= \arg\max_{u\in\mathbb{R}^n} \rho^y(u) = \arg\min_{u\in\mathbb{R}^n} (u-m)^\top C^{-1}(u-m) = m, \\ u_{\mathrm{CM}} &= \mathbb{E}(u|y) = m, \\ u_{\mathrm{opt}} &= \arg\min_{u\in\mathbb{R}^n} \|Ax-y\|_\Gamma^2 + \|u-m_0\|_{C_0}^2 = m. \end{aligned}$$

Example 2.3 (Comparison of CM & MAP (1D))**.** *Let u be a real-valued rv and assume that the posterior density is given by*

$$\rho^y(u) = \frac{\alpha}{\sigma_0}\varphi\left(\frac{u}{\sigma_0}\right) + \frac{1-\alpha}{\sigma_1}\varphi\left(\frac{u-1}{\sigma_1}\right) \text{ with } 0<\alpha<1,\ \sigma_0,\sigma_1>0 \text{ and } \varphi(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right).$$

Estimators.

$$u_{\mathrm{CM}} = 1-\alpha,\ u_{\mathrm{MAP}} = \begin{cases} 0, & \text{if } \frac{\alpha}{\sigma_0} > \frac{1-\alpha}{\sigma_1}, \\ 1, & \text{if } \frac{\alpha}{\sigma_0} < \frac{1-\alpha}{\sigma_1}, \\ \{0,1\}, & \text{if } \frac{\alpha}{\sigma_0} = \frac{1-\alpha}{\sigma_1}. \end{cases}$$

A Tools

A.1 Probability Theory

A.1.1 Background in Probability Theory

Probability Space $(\Omega, \mathcal{A}, \mu)$

- Ω is a set
- $\mathcal{A}$ is a σ-algebra over Ω
  - $\emptyset \in \mathcal{A}$,
  - the complement $A^c := \{\omega\in\Omega; \omega\notin A\} \in \mathcal{A}$ for all $A\in\mathcal{A}$ and
  - the union $\bigcup_{j\in\mathbb{N}} A_j \in \mathcal{A}$ for every sequence $(A_j)_{j\in\mathbb{N}} \in \mathcal{A}$.

- A *measure* μ on a measurable space $(\Omega, \mathcal{A})$ is a mapping from $\mathcal{A}$ to $\mathbb{R}_+ \cup \{\infty\}$ s.t.
  - the empty set has measure zero, i.e., $\mu(\emptyset) = 0$ and
  - $\mu\left(\bigcup_{j\in\mathbb{N}} A_j\right) = \sum_{j\in\mathbb{N}} \mu(A_j)$ if $A_j \in \mathcal{A}$ are disjoint.
- μ is a *probability measure* iff
  - $\mu(\Omega) = 1$
- $\mathcal{B}(\mathbb{R})$ denotes the *Borel σ-algebra* and equals the smallest σ-algebra containing all open subsets of $\mathbb{R}$.
- The usual notion of volume of subsets of $\mathbb{R}^d$ gives rise to the *Lebesgue measure*.

A.1.2 Real-valued Random Variables

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space.

A mapping $f : \Omega \to \mathbb{R}$ is called a *random variable*, if f is $\mathcal{A}$-$\mathcal{B}(\mathbb{R})$-measurable, i.e.,

$$\forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) = \{\omega \in \Omega; f(\omega) \in B\} \in \mathcal{A} \Rightarrow f^{-1}(\mathcal{B}(\mathbb{R})) \subset \mathcal{A}.$$

- Image measure/distribution
  - $\mu_f(B) := (\mu \circ f^{-1}(B)) = \mu(f^{-1}(B)) = \mu(\{f \in B\})$.
  - $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_f)$ is again a probability measure.
- Distribution function
  - $F_f(x) := \mu_f((-\infty, x]) = \mu(f \le x) = \mu(\{\omega \in \Omega; f(\omega) \le x\})$.
- Density function
  - If $\mu_f(B) := \int_B \rho(x)\mathrm{d}x$ for all $B \in \mathcal{B}(\mathbb{R})$ with a measurable function $\rho : \mathbb{R} \to [0, \infty)$.
- If h is a real-valued measurable function on $\mathbb{R} \Rightarrow h \circ f$ $\mathcal{A}$-$\mathcal{B}(\mathbb{R})$-measurable

A.1.3 Expectations

- For $f \in \mathcal{L}^1(\mu)$ is the *expectation* defined as

$$\mathbb{E}(f) = \int_\Omega f \mathrm{d}\mu = \int_\Omega f(\omega)\mathrm{d}\mu(\omega).$$

- For a real-valued measurable function h
  - discretely distributed random variable: $\mu_f = \sum_{i=1}^n p_i \mathbf{1}_{x_i}$

$$\mathbb{E}(h \circ f) = \int_\Omega h \circ f \mathrm{d}\mu = \int_\mathbb{R} h \mathrm{d}\mu_f = \sum_{i=1}^n p_i f_{x_i}.???$$

  - continuously distributed random variable: $\mu_f([a, b]) = \int_a^b \rho(x)\mathrm{d}x$.

$$\mathbb{E}(h \circ f) = \int_\Omega h \circ f \mathrm{d}\mu = \int_\mathbb{R} h \mathrm{d}\mu_f = \int_\mathbb{R} h(x)\rho(x)\mathrm{d}x.$$

- For $f \in \mathcal{L}^2(\mu)$ is the *variance* defined as

$$\mathbb{V}(f) = \mathbb{E}((f - \mathbb{E}(f))^2) = \mathbb{E}(f^2) - (\mathbb{E}(f))^2.$$

A.1.4 Product Spaces & Joint Distributions

- Product space $(\bigtimes_{i=1}^N \Omega_i, \bigotimes_{i=1}^N \mathcal{A}_i, \bigotimes_{i=1}^N \mu_i)$
  - $\bigtimes_{i=1}^N \Omega_i = \{(\omega_1, \dots, \omega_N); \omega_i \in \Omega_i,\ i = 1, \dots, N\}$.
  - $\bigotimes_{i=1}^N \mathcal{A}_i$ smallest σ-algebra that contains all sets of the form $A = A_1 \times \cdots \times A_N$ with $A_i \in \mathcal{A}_i$, $i = 1, \dots, N$.
  - $\bigotimes_{i=1}^N \mu_i$ is defined by

$$\bigotimes_{i=1}^N \mu_i = \mu_1(A_1) \times \cdots \times \mu_N(A_N),\ \forall A_1 \in \mathcal{A}_1, \dots, A_N \in \mathcal{A}_N.$$

- For a real-valued random variables $(f_i,\ i = 1, \dots, N)$, we define $f_1 \otimes \cdots \otimes f_N : \Omega \to \mathbb{R}^N$ via $\omega \mapsto (f_1(\omega), \dots, f_N(\omega))$ a random variable in $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N))$.
  - Joint distribution is for $B \in \mathcal{B}(\mathbb{R}^N)$ given by

$$\mu_{f_1 \otimes \cdots \otimes f_N}(B) = \mu(f_1 \otimes \cdots \otimes f_N \in B).$$

  - Joint distribution function is given by

$$F_{f_1 \otimes \cdots \otimes f_N}(x_1, \dots, x_N) = \mu_{f_1 \otimes \cdots \otimes f_N}((-\infty, x_1], \dots, (-\infty, x_N]) = \mu(f_1 \le x_1, \dots, f_n \le x_n).$$

A.1.5 Conditional Probabilities & Expectations

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space.

- The *conditional probability* that an event $A \in \mathcal{A}$ occurs given that an event $B \in \mathcal{A}$ has occurred is defined by

$$\mu(A|B) = \frac{\mu(A \cap B)}{\mu(B)} \text{ for } \mu(B) > 0.$$

- Given 2 random variables f and h with joint density function ρ and $x \in \mathbb{R}$ s.t. $\int_{-\infty}^{\infty} \rho(x, v)\mathrm{d}v > 0$,
  - the *conditional distribution function* of h given $f = x$ is given by

$$F_{h|f}(y|x) = \int_{-\infty}^{y} \frac{\rho(x, v)\mathrm{d}v}{\int_{-\infty}^{\infty} \rho(x, v)\mathrm{d}v}.$$

  - the *conditional density function* of h given $f = x$ is given by

$$\rho_{h|f}(y|x) = \frac{\rho(x, y)}{\int_{-\infty}^{\infty} \rho(x, v)\mathrm{d}v}.$$

  - the *conditional expectation* $\mathbb{E}(h|f)$ is defined by

$$\mathbb{E}(h|f) = \int_{-\infty}^{\infty} y \rho_{h|f}(y|x)\mathrm{d}y.$$

A.1.6 Independent Random Variables

- For *independent random variables* $(f_i)_{i=1}^N$, it holds

$$\begin{aligned}\mu_{f_1 \otimes \cdots \otimes f_N}(B_1 \times \cdots \times B_N) &= \mu(f_1 \in B_1, \dots, f_N \in B_N) = \mu(f_1 \in B_1) \cdots \mu(f_N \in B_N) \\ &= \mu_{f_1}(B_1) \cdots \mu_{f_N}(B_N) = \mu_{f_1} \otimes \cdots \otimes \mu_{f_N}(B_1 \times \cdots \times B_N).\end{aligned}$$

- $(f_i)_{i=1}^N$ are independent $\Leftrightarrow$ the joint distribution is equal to the product measure.

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